

NUMBER OF INCREASING SEQUENCES IN A PERMUTATION

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We investigate the expected number of increasing subsequences in a sequence of values. Specifically, given n points $x_1 < x_2 < \dots < x_n$ uniformly distributed in the interval $(0, 1)$, we apply the mapping $\sigma(i) = ax_i \mod 1$, where a is a fixed integer. We then analyze the behavior of the number of increasing subsequences in the transformed sequence as both a and n grow large.

An *increasing subsequence* is a subset of elements extracted from a given sequence while preserving their relative order and ensuring that each subsequent element is strictly greater than the preceding one.

The study of increasing subsequences plays a fundamental role in combinatorial mathematics and probability theory, as it reveals structural properties and order patterns within randomly generated sequences. As an illustrative example, consider a sequence of values x_i uniformly distributed in the interval $(0, 1)$, arranged in increasing order, meaning that

$$x_1 < x_2 < \dots < x_n.$$

We then define a mapping σ on the set $\{1, 2, \dots, n\}$ by $\sigma(i) = ax_i \mod 1$, where a is a fixed integer. Our primary objective is to analyze the expected number of increasing subsequences when both a and n are sufficiently large.

This problem finds a significant application in the well-known card shuffling technique called the *Riffle Shuffle*. Specifically, an a -Riffle Shuffle consists of splitting a deck of n cards into a sub-decks and then interleaving them in a random fashion while preserving the relative order of the cards within each sub-deck. When $a = 2$, this is known as a 2-Riffle Shuffle or simply a Riffle Shuffle. If the n cards are initially numbered from 1 to n in increasing order, the original deck consists of a single increasing sequence. After performing an a -Riffle Shuffle, at most a increasing subsequences are created. This leads to the central question of our study: What is the expected number of increasing subsequences when a and n are sufficiently large? Answering this question provides valuable insights into the structure of shuffled sequences and their probabilistic properties.

In [1], Brad Mann claims, without providing a proof, that after an a -riffle shuffle, the expected number of rising sequences is given by the following expression:

$$a - \frac{n+1}{a^n} \sum_{r=1}^{a-1} r^n.$$

We provide a proof here that the expected number of rising sequences is actually given by:

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$$(1) \quad R_{a,n} = 1 + (a-1) \left(1 - \left(1 - \frac{1}{a} \right)^n \right) - \frac{n(a-1)^n}{a^n} - \frac{1}{a^n} \sum_{i=1}^{a-2} \frac{(a-1)^n - i^n}{a-1-i}.$$

This expression makes sense because when n is sufficiently large than a and a is also large, we observe that $R_{a,n}$ converges to a , which represents the maximum number of rising sequences. Furthermore, as mentioned in [1], the expected number of rising sequences satisfies the asymptotic behavior:

$$R_{a,n} = O(n).$$

Proof. Let $a, n \in \mathbb{N}$ such that $a < n$. We define X as a random variable taking values in $\mathfrak{S}_n$, representing the outcome of an a -riffle shuffle. Here, we consider the inverse version of the a -riffle shuffle, which is equivalent to the standard a -riffle shuffle but viewed in reverse. The a -inverse riffle shuffle reconstructs the hidden structure of the shuffle by reassigning each card an original label corresponding to one of the a different subdivision blocks before the interleaving step of the riffle shuffle. This process can be described as follows: each card is directly assigned an integer from the set $\{p_1, p_2, \dots, p_a\}$, representing its membership in a specific sub-packet. The labels p_i 's are integers such that $p_i < p_{i+1}$.

To reverse the shuffle, we start with the bottommost card, check its label, and place it in the corresponding block. This process continues sequentially up to the topmost card. The sub-packets are then stacked in ascending order of their labels: the cards from sub-packet p_1 are placed at the bottom, those from p_2 above them, and so on, until the cards from p_a form the topmost layer.

The advantage of using the inverse riffle shuffle is that it allows us to describe how new rising sequences are formed more easily. Specifically, a new rising sequence appears whenever the last card labeled p_i is placed after at least one card with a strictly greater label in the new formed block.

We define the function

$$\begin{aligned} f : X(\Omega) &\longrightarrow \mathbb{R} \\ \pi &\longmapsto f(\pi), \end{aligned}$$

where $f(\pi)$ denotes the number of rising sequences in the permutation π .

The expected number of rising sequence in a permutation coming from an a -Riffle Shuffle is then given by $\mathbb{E}[f(X)]$.

For each $i \in \{1, \dots, a-1\}$, we define the event

$$\begin{aligned} A_i &= \{\pi \in X(\Omega) \mid \text{the last card labeled } i \text{ appears after at least one card labeled } j \geq i+1\} \\ &= \{\pi \in \mathfrak{S}_n \mid \text{the last card labeled } i \text{ appears after at least one card labeled } j \geq i+1\}, \end{aligned}$$

since $X(\Omega) = \mathfrak{S}_n$.

The number of rising sequences in π can be expressed as:

$$f(\pi) = 1 + \sum_{i=1}^{a-1} \mathbb{1}_{A_i}(\pi).$$

The additional 1 in the expression for $f(\pi)$ accounts for the fact that there is always at least one rising sequence initiated by the first card.

The expectation is then given by:

$$\begin{aligned}\mathbb{E}[f(X)] &= \mathbb{E}\left[1 + \sum_{i=1}^{a-1} \mathbf{1}_{A_i}(X)\right] \\ &= 1 + \sum_{i=1}^{a-1} \mathbb{E}(\mathbf{1}_{X \in A_i}) = 1 + \sum_{i=1}^{a-1} \mathbb{P}(A_i)\end{aligned}$$

To better understand how $\mathbb{P}(A_i)$ is computed for each i , we introduce an equivalent problem with a more intuitive formulation.

Consider a pile of n socks, each randomly labeled with a number from the set $\{1, \dots, a\}$, where $a < n$. These socks are stacked in a completely random order. The sorting process is as follows: at each step, we take the sock at the top of the pile, read its label, and place it in the corresponding drawer, numbered from 1 to a . The total number of ways to arrange these n socks into a drawers is a^n .

Now, given a label $i \in \{1, \dots, a\}$, we seek to determine how many of these arrangements satisfy the condition that the last sock with label i is placed after at least one sock with a label in $\{i+1, \dots, a\}$.

To answer this question, we consider the negation: this means that before the last sock with label i , no sock has a number greater than i . If the last sock labeled i is in position j , then for the first $j-1$ socks, **there are i^{j-1} possible ways to assign labels (choosing values less than or equal to i)**. The sock in position j must have label i , after which the remaining $n-j$ socks can take any label except i , leading to $(a-1)^{n-j}$ possibilities. Additionally, we must consider the case where no sock is labeled i , which contributes $(a-1)^n$ possibilities.

Thus, the total number of arrangements where the last sock labeled i appears before any sock with a greater label is:

$$a^n - (a-1)^n - \sum_{j=1}^n i^{j-1} (a-1)^{n-j}.$$

Therefore, the probability that the last sock labeled i appears after at least one sock with a greater label is:

$$\mathbb{P}(A_i) = \frac{1}{a^n} \left(a^n - (a-1)^n - \sum_{j=1}^n i^{j-1} (a-1)^{n-j} \right).$$

Returning to our original problem, we obtain:

$$\mathbb{E}[f(X)] = 1 + (a-1) \left(1 - \left(1 - \frac{1}{a} \right)^n \right) - \frac{n(a-1)^n}{a^n} - \frac{1}{a^n} \sum_{i=1}^{a-2} \frac{(a-1)^n - i^n}{a-1-i}.$$

This final expression provides the expected number of rising sequences after an a -riffle shuffle.

One could also notice that

$$\sum_{i=1}^{a-2} \frac{(a-1)^n - i^n}{a-1-i} = (a-1)^{n-1} \sum_{i=1}^{a-2} \frac{1 - \left(\frac{i}{a-1}\right)^n}{1 - \frac{i}{a-1}}.$$

Setting $x_i = i/(a-1)$ and for sufficiently large n , we have $x^n \rightarrow 0$, so that

$$\sum_{i=1}^{a-2} \frac{(a-1)^n - i^n}{a-1-i} \approx (a-1)^{n-1} \sum_{i=1}^{a-2} \frac{1}{1-x_i}.$$

Setting $i = a-1-j$, we find that $x_i = 1-x_j$, hence

$$\sum_{i=1}^{a-2} \frac{1}{1-x_i} = \sum_{j=1}^{a-2} \frac{1}{x_j} \sim (a-1) \log(a).$$

Therefore,

$$\sum_{i=1}^{a-2} \frac{(a-1)^n - i^n}{a-1-i} \sim (a-1)^n \log(a).$$

Letting $\theta_n := \left(1 - \frac{1}{a}\right)^n$, we obtain:

$$R_{a,n} \approx 1 + (a-1)(1-\theta_n) - n\theta_n - \theta_n \log a,$$

so that

$$R_{a,n} \approx a - \theta_n ((a-1) + n + \log a).$$

Thus,

$$R_{a,n} \approx a - (a + n + \log a) e^{-\frac{n}{a}}.$$

REFERENCES

- [1] Brad Mann. HOW MANY TIMES SHOULD YOU SHUFFLE A DECK OF CARDS?1, 1998,
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